

Mazen Ahmed Shams

Faculty of Engineering, Alexandria University
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Age: 24 | Military Status: Exempted

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EDUCATION

Faculty of Engineering, Alexandria University
Bachelor of Engineering, Major in Computer Science and Communication Engineering.

October 2019 – July 2024
CGPA: 3.17

PROFESSIONAL EXPERIENCE

- Prometeon Tyres Group - Smart Manufacturing Engineer**
November 2025 – Present
- HEMS (Smart Asset Management):** System designed and deployed a FastAPI/HTMX/Tailwind web app (using AI code orchestration) to track 400+ handling assets, migrating manual workflows to a secure QR-to-repair system.
 - Thermal Asset Monitoring:** Engineered a preventative maintenance Python pipeline extracting thermal imaging data from FLIR cameras to visualize asset health on Grafana.
 - IIoT Network Integration:** Configured Kepware/ThingWorx to route Siemens/Rockwell PLC data, and built a Java middleware POC using the ThingWorx Edge SDK (AlwaysOn WebSocket protocol) for real-time streaming.
 - AI Troubleshooting (Product Owner):** Conceptualized a local RAG troubleshooter (Streamlit/ChromaDB/Ollama) for troubleshooting manuals; gathered requirements from maintenance teams and collaborated with IT to validate diagnostics.
- DEPI - Microsoft Cloud Data Engineer Training Program**
June 2025 – November 2025
- Cloud Data Architecture:** Completed intensive training on the Microsoft Azure Data ecosystem, mastering data management, processing, and Cloud Data Engineering.
 - Advanced Implementations:** Designed a SQL Data Warehouse and utilized advanced Python to build scalable data solutions, culminating in a DevOps-aligned capstone project.
- iSchool - Software Coding Instructor**
December 2024 – June 2025
- Technical Instruction:** Delivered engaging, project-based lessons on coding fundamentals and problem-solving to students aged 8 to 16.
- Faculty of Engineering Alexandria University - Teaching Assistant (TA)**
Sept. 2024 – Dec. 2024
- Course Instruction:** Taught Programming Language Translation (PLT), assisting undergraduate students with compiler concepts and optimizing academic projects.

TECHNICAL SKILLS

Languages: Python, SQL (T-SQL), C/C++, Java, Golang

Cloud: Microsoft Azure (Data Factory, Stream Analytics, Event Hubs), AWS Cloud Computing

Data Engineering: Apache Kafka, Batch ETL, Dremio

Databases: Azure SQL, MS SQL Server, MySQL, Snowflake

Visualization: Power BI (DAX), Grafana

IIoT & Automation: Kepware, ThingWorx, PLC Data Integration, FLIR Thermal Imaging, Microcontrollers (Arduino)

Tools: pandas, SQLAlchemy, Docker, Git, MS Copilot

SELECTED PROJECTS

More projects on my GitHub (Click here).

- Hybrid IoT Data Pipeline:** Scalable pipeline simulating sensors, using Kafka, Azure Event Hubs, and Power BI.
- Retail Data Pipeline:** Prototyped batch ingest-and-transform ETL using modular notebooks for reproducible runs.
- Graduation Project — ASD App:** Mobile app for autistic children with AI chatbot, AR, and emotion recognition.

LICENSES AND CERTIFICATES

- Microsoft:** AI Skills Fest (2026)
- Udacity:** AWS AI Practitioner Challenge (2026)
- DataCamp:** Certified Data Engineer (2025) | Associate Data Engineer
- AWS Academy Graduate:** Data Analytics, Machine Learning Foundations, Cloud Foundations